
DECODING CROWDFUNDING SUCCESS

A STATISTICAL JOURNEY THROUGH KICKSTARTER DATA

*A Dual-Model Analysis Using Logistic & Zero-Inflated
Poisson Regressions*

PRIYASMITH GHOSH

August 2026

DECLARATION

I hereby declare that this report, titled "**Decoding Crowdfunding Success: A Statistical Journey Through Kickstarter Data**", is an original, independent statistical project authored by me.

All predictive models, data engineering pipelines, and statistical analyses presented in this document were developed independently. Where open-source software, external datasets, or theoretical frameworks have been utilized, they have been appropriately referenced and acknowledged. This work has not been submitted for the award of any academic degree or diploma at any university or institution.

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August 2026

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Finally, I acknowledge the public availability of the Kickstarter dataset, which provided the rich, real-world data necessary to test and validate the Generalized Linear Models presented in this work.

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ABSTRACT

Crowdfunding platforms like Kickstarter have democratized early-stage venture capital, yet the mechanics of campaign success remain highly volatile and mathematically complex. This study presents a comprehensive statistical pipeline to decode these dynamics through a dual-model Generalized Linear Modeling (GLM) framework. First, a Logistic Regression architecture is deployed to classify binary campaign outcomes, utilizing strategic data engineering and ROC/AUC metrics to isolate the strongest predictors of funding success. Second, to address the extreme variance and inherent structural zeros present in backer volumes, a Zero-Inflated Poisson (ZIP) regression model is constructed. By formally testing for and mitigating overdispersion, this dual approach moves beyond standard binary classification to accurately model overdispersed count data. Ultimately, this analysis bridges the gap between predictive data science and theoretical statistical inference, offering a rigorous econometric blueprint of crowdfunding behavior.

INTRODUCTION

"Crowdfunding represents a paradigm shift in how early-stage ventures secure capital, relying on the collective financial power of the public rather than traditional institutional investors."

Over the past decade, reward-based platforms like Kickstarter have democratized the financial landscape. However, this decentralized model introduces extreme volatility. Despite billions of dollars in pledged funds, a significant percentage of campaigns fail to reach their strict "all-or-nothing" targets. For creators and analysts, decoding the mechanics of campaign success requires moving beyond anecdotal advice and applying rigorous statistical modeling to objective features.

This project tackles the Kickstarter ecosystem as a two-fold statistical challenge within the framework of **Generalized Linear Models (GLMs)**. The first challenge is *binary classification*: a campaign is definitively successful or failed. To model this, a Logistic Regression architecture is deployed, utilizing a logit link function to estimate the log-odds of success and evaluating predictive power via Receiver Operating Characteristic (ROC) and Area Under the Curve (AUC) metrics.

However, evaluating crowdfunding purely through a binary lens ignores the volumetrics of crowd participation. The true scale of a campaign is driven by its total backers, necessitating *count data analysis*. Standard Poisson regression fails here due to severe overdispersion and a massive influx of structural zeros—countless projects launch into the void and receive exactly zero backers. To resolve this, a Zero-Inflated Poisson (ZIP) regression model is implemented. This dual-process architecture explicitly accounts for excess zeros by blending a logistic process (predicting structural zeros) with a Poisson process (modeling overdispersed counts), providing a robust solution for highly skewed financial data.

This report provides a comprehensive econometric blueprint of crowdfunding behavior. Subsequent chapters detail the data engineering pipelines, the theoretical frameworks governing zero-inflation and overdispersion, and the diagnostic evaluation of both the Logistic and ZIP models, ultimately culminating in actionable strategic insights.

Chapter 1

DATA OVERVIEW & PREPROCESSING

1.1 The Kickstarter Dataset

The empirical foundation of this analysis is built upon a publicly available Kickstarter dataset sourced from Kaggle :

<https://www.kaggle.com/datasets/kemical/kickstarter-projects>.

The dataset aggregates historical crowdfunding campaigns spanning from the platform’s inception through 2018, providing a multidimensional view of project characteristics across hundreds of thousands of observations. This extensive timeline and massive sample size supply the necessary statistical power to train robust classification and count-data models, ensuring the resulting inferences are stable and broadly applicable to the crowdfunding ecosystem.

1.2 Raw Data Snapshot

To provide a concrete visualization of the input structure, Table 1.1 displays a representative sample of raw records extracted directly from the Kickstarter dataset prior to preprocessing and filtering.

Table 1.1: Sample Snapshot of Raw Kickstarter Records

ID	main_category	usd_goal_real	state	backers	duration_days
1000002330	Publishing	1000.00	failed	0	59.0
1000003930	Film & Video	30000.00	failed	15	30.0
1000004038	Film & Video	45000.00	failed	3	60.0
1000007540	Music	5000.00	successful	224	30.0

1.3 Data Dictionary and Column Explanations

To facilitate the dual-model architecture of this study, specific variables were isolated from the raw data dictionary to serve as response targets and structural predictors. Table 1.2 summarizes the core raw variables utilized to engineer our final modeling matrix.

Table 1.2: Data Dictionary of Key Raw Kickstarter Variables

Variable Name	Description / Operational Definition	Role in Study
state	Final status of the campaign (e.g., successful, failed, canceled) .	Binary Response (Logistic)
backers	Total count of individual financial contributors to the project .	Count Response (ZIP)
usd_goal_real	Funding goal standardized into US Dollars (USD) via API conversion .	Continuous Predictor
main_category	Overarching industry classification (e.g., Technology, Art, Games) .	Categorical Predictor
category	Granular sub-category classification (e.g., Hardware, Web, Poetry).	Micro-Market Predictor
launched / deadline	Exact timestamp of when the campaign went live and its funding cutoff.	Temporal Predictors
NAME	The official title of the campaign provided by the creator.	Marketing Predictor
country	Two-letter country code of the campaign's origin.	Geographic Predictor

1.4 Data Cleaning and Preprocessing

Real-world financial data is inherently noisy and requires rigorous preprocessing to ensure model validity. The data engineering pipeline consisted of three primary phases:

1.4.1 State Filtration

The raw dataset contained campaigns in transitional states, such as 'live' and 'undefined' . Because the objective of this study is to analyze definitive outcomes, these intermediate observations were programmatically filtered out . The states 'canceled' and 'suspended' were treated as the Failed category . By this, the dataset was strictly con-

strained to campaigns whose state resolved to either 'successful' or 'failed', establishing a clean binary outcome space .

1.4.2 Variable Selection for *A Priori* Prediction

To build an honest, pre-launch predictive model, we must extract maximum informational signal from the data available *before* the campaign begins to accumulate backers. Relying solely on the broad category and financial goal leaves the model highly susceptible to Omitted Variable Bias. To capture this unobserved heterogeneity, several new *a priori* features were engineered from the raw columns:

- **Campaign Duration** (`duration_days`): The original dataset provided launched and deadline timestamps . A continuous variable was engineered by calculating the exact time (in days) between these two points, operationalizing the time constraint imposed on the creator .
- **Seasonality** (`launch_month`): Crowdfunding is subject to cyclical consumer spending habits. The month was extracted from the launched timestamp to serve as a categorical predictor, accounting for seasonal fatigue (e.g., holiday periods).
- **Marketing Clarity** (`name_length`): The character count of the campaign's NAME was calculated. This serves as a quantifiable proxy for the creator's marketing effort and the psychological clarity of their pitch.
- **Geographic Trust** (`is_US`): A binary indicator was created from the country column, isolating US-based campaigns (1) from international ones (0) to account for baseline platform trust and domestic market size advantages.

These engineered variables provide a highly dimensional, pre-treatment covariate matrix, ensuring the Generalized Linear Models can detect nuanced micro-market dynamics without relying on post-launch data leakage.

1.4.3 Logarithmic Transformations

Financial metrics, particularly `usd_goal_real`, exhibited severe positive skewness characterized by an elongated right tail, driven by a small subset of ambitious campaigns seeking extraordinary funding amounts. In raw financial data, this extreme asymmetry can disproportionately bias parameter estimates, inflate standard errors, and severely degrade predictive accuracy. To stabilize the residual variance, and ensure the numerical stability and rapid convergence of the Maximum Likelihood Estimation

(MLE) algorithms governing our Generalized Linear Models, a logarithmic transformation was applied:

$$\log_usd_goal = \log(usd_goal_real + 1) \quad (1.1)$$

The inclusion of the additive offset term (+1) acts as a structural safeguard, preventing undefined logarithmic values ($-\infty$) in the rare scenario of a *zero-dollar funding goal configuration*. By compressing the scale of extreme values, this transformation normalizes the distribution of the covariate, linearizes potential non-linear relationships with the response variables, and yields a far more robust foundation for statistical inference.

THEORETICAL FRAMEWORK

2.1 Basic Terminologies

2.1.1 Structural Zeros vs. Sampling Zeros

In count data modeling, a *structural zero* occurs when a zero outcome is inevitable due to the underlying mechanics of the observation (e.g., a campaign that was utterly abandoned upon launch). A *sampling zero* occurs by chance in a standard Poisson process. Distinguishing between the two is critical for model selection.

2.1.2 Overdispersion

Overdispersion occurs when the observed variance in a dataset exceeds the theoretical variance assumed by a specified model, a common violation in standard Poisson regression where the mean and variance are assumed equal.

2.2 Generalized Linear Models (GLMs)

GLMs extend classical Ordinary Least Squares (OLS) regression by allowing the linear model to be related to the response variable via a *link function*, accommodating error distributions other than the Normal distribution (such as Binomial or Poisson).

2.3 Logistic Regression for Binary Classification

2.3.1 Model Formulation and Logit Link

Let $Y_i \sim \text{Bernoulli}(p_i)$. The logistic regression model uses the logit link function to model the probability of campaign success:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_k X_{ki} \quad (2.1)$$

2.3.2 Interpretation of Coefficients (Log-Odds)

The exponentiated coefficients, $\exp(\beta_j)$, represent the odds ratios. A value greater than 1 indicates that an increase in the predictor increases the odds of campaign success.

2.4 Zero-Inflated Poisson (ZIP) Regression

While logistic regression handles binary outcomes, modeling the sheer volume of crowd participation requires count data methodologies.

2.4.1 Addressing Overdispersed Count Data

Standard Poisson regression fails when the dataset exhibits an excessive number of zeros (campaigns with 0 backers). The ZIP model addresses this by assuming the data is generated by two distinct processes.

2.4.2 Population Heterogeneity and the Spike at Zero

Count data modeling often involves underlying populations that are not homogeneous. Common examples of this phenomenon include the number of cigarettes smoked daily, the number of glasses of alcohol consumed in a month, or the number of movies watched monthly on Netflix. When plotting a histogram of such data, the visualization will typically display a distinct spike at zero.

In the context of Kickstarter, this heterogeneity represents the divide between campaigns that launch and are dead on arrival (*guaranteeing zero backers*) versus those that gain traction.

2.4.3 The ZIP Model Architecture

To mathematically accommodate this phenomenon, we deploy a Zero-Inflated Poisson (ZIP) model. The ZIP model probability mass function is defined as a mixture:

$$f(y|\mathbf{x}, \mathbf{z}, \boldsymbol{\beta}, \gamma) = f_z(0; \mathbf{z}, \gamma) \cdot I_{\{0\}}(y) + (1 - f_z(0; \mathbf{z}, \gamma))f_p(y; \mathbf{x}, \boldsymbol{\beta}) \quad (2.2)$$

where:

- $I_{\{0\}}(y)$ represents the indicator function.
- $\pi = P[y = 0|\mathbf{z}, \gamma] = g^{-1}(\mathbf{z}'\boldsymbol{\gamma})$ is the probability of observing a structural zero, meaning the zero count is inflated with probability π .
- $f_p(y; \mathbf{x}, \boldsymbol{\beta})$ represents any standard count distribution (in this case, the Poisson distribution).

A defining characteristic of the ZIP model is that the zero count originates from both distributions—the zero-inflation component and the standard count component. This architecture allows us to simultaneously model the barrier to entry (getting the first backer) and the scale of success (accumulating total backers).

Note: While Hurdle models share a similar conceptual foundation to ZIP models, they operate using a fundamentally different parametrization, making the ZIP architecture our preferred choice for this specific application.

2.5 Model Evaluation Metrics

To ensure the statistical validity and predictive power of the applied models, rigorous diagnostic metrics are required. Because this study employs both classification (Logistic) and count-data (ZIP) methodologies, the evaluation framework must be partitioned accordingly.

2.5.1 Classification Accuracy and the Confusion Matrix

For the Logistic Regression model, predictive performance at a definitive cutoff threshold (typically 0.5) is evaluated using a Confusion Matrix. This matrix cross-tabulates the empirically observed outcomes against the model's predicted classifications, isolating the exact nature of the prediction errors.

Table 2.1: Structure of the Binary Confusion Matrix

	Predicted Success (1)	Predicted Failure (0)
Actual Success (1)	True Positive (TP)	False Negative (FN)
Actual Failure (0)	False Positive (FP)	True Negative (TN)

From this matrix, the primary metric of **Accuracy** is derived, representing the proportion of total predictions that were fundamentally correct:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2.3)$$

2.5.2 Sensitivity and Specificity

While highly interpretable, global accuracy can be misleading in imbalanced datasets. To rigorously evaluate the model, we follow the theoretical framework established in *Categorical Data Analysis*. It defines the intrinsic accuracy of a classifier using conditional probabilities, completely independent of the underlying class prevalence.

Let Y denote the true empirical state ($Y = 1$ for a successful campaign, $Y = 0$ for a failed campaign) and let $\hat{Y}$ denote the model's predicted classification.

- **Sensitivity (True Positive Rate):** The probability that the model correctly predicts a successful outcome, given that the campaign was actually successful.

$$\text{Sensitivity} = P(\hat{Y} = 1 \mid Y = 1) = \frac{TP}{TP + FN} \quad (2.4)$$

- **Specificity (True Negative Rate):** The probability that the model correctly predicts a failed outcome, given that the campaign actually failed.

$$\text{Specificity} = P(\hat{Y} = 0 \mid Y = 0) = \frac{TN}{TN + FP} \quad (2.5)$$

While sensitivity focuses strictly on correctly classifying the "successes," specificity is required to evaluate the accurate classification of the "failures." Both conditional metrics must be analyzed in tandem to fully evaluate the discriminative power of the binary classifier.

2.5.3 ROC Curve and AUC: What, When, and Why

To evaluate the discriminative capacity of the logistic model across *all* possible classification thresholds, we utilize the **Receiver Operating Characteristic (ROC) curve**.

- **What is it?**

The ROC curve is a graphical plot that contrasts the True Positive Rate (Sensitivity) against the False Positive Rate ($1 - \text{Specificity}$) as the probability threshold for classification varies from 0 to 1.

$$\text{TPR} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN} \quad (2.6)$$

The **Area Under the Curve (AUC)** is the scalar integral of the ROC curve. An AUC of 0.5 denotes a model with no discriminatory power (equivalent to a coin toss), while an AUC of 1.0 signifies perfect classification.

- **Why it is used:** The ROC/AUC framework isolates the model's underlying mathematical ability to rank a randomly chosen successful campaign higher than a randomly chosen failed campaign, completely independent of the arbitrary 0.5 cutoff point.
- **When it is applied:** It is the gold standard for evaluating binary classifiers in financial and econometric contexts, especially when the cost of a False Positive (e.g., investing in a doomed campaign) differs significantly from the cost of a False Negative (e.g., passing on a highly profitable venture).

2.5.4 Count Model Diagnostics (ZIP)

Standard classification metrics are fundamentally incompatible with count data. To validate the Zero-Inflated Poisson model, we rely on distributional diagnostics and formal econometric tests.

The Zero-Catch Metric

The primary objective of implementing a ZIP architecture is to resolve the overdispersion caused by structural zeros. The "Zero-Catch" metric empirically validates this by comparing the exact number of zeros predicted by the model against the actual number of zeros present in the raw data. A standard Poisson model will characteristically under-predict zeros in Kickstarter data; a successful ZIP model should perfectly align with the empirical zero-mass.

EXPLORATORY DATA ANALYSIS & VISUAL DIAGNOSTICS

Exploratory Data Analysis (EDA) is not merely a descriptive exercise; it is a critical diagnostic step that dictates the mathematical architecture of the subsequent predictive models. In this chapter, we visually and quantitatively investigate the distributional geometries of the Kickstarter dataset, explicitly detailing the transition from raw, noisy data to the stabilized, cleaned inputs required for Generalized Linear Modeling.

3.1 Data Filtration and the Binary Response Space

The primary objective of the classification model is to predict campaign success. However, analyzing the raw data reveals a fractured outcome space that requires rigorous consolidation to formulate a valid Bernoulli response variable.

3.1.1 The Raw State (Before Cleaning)

As illustrated in Figure 3.1, the raw Kickstarter dataset extracted from the API contains campaigns distributed across six distinct operational states: *canceled*, *failed*, *live*, *successful*, *suspended*, and *undefined*.

To investigate this initial distribution, the following R code was utilized to visualize the raw states prior to any filtration:

```
1 # Checking the raw distribution of campaign states
2 table(data$state)
3
4 # Plotting the raw bar plot
5 ggplot(data, aes(x = state)) +
6   geom_bar(fill = "indianred", colour="black") +
7   geom_text(stat="count", aes(label=after_stat(count)), vjust=-0.5, size
8     = 3.5,
9     fontface= "bold")+
10
```

```

9  labs(title = "RAW CAMPAIGN STATES (BEFORE CLEANING)", x = "State", y
    = "Count") +
10  theme_gray() +
11  theme(
12    plot.title = element_text(hjust=0.5, face="bold", size= 14)
13  ) +
14  ylim(0, 210000)

```

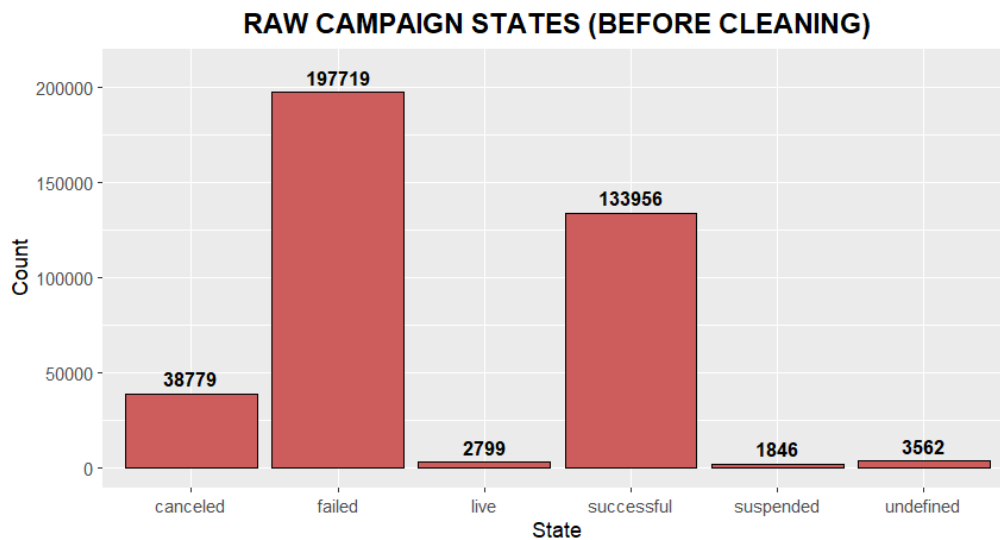


Figure 3.1: Frequency distribution of raw campaign states prior to data filtration.

The presence of intermediate and aborted states (e.g., 2,799 ‘live’ campaigns and 3,662 ‘undefined’ campaigns) introduces severe censorship and ambiguity into the predictive timeline. A ‘live’ campaign is right-censored—its true final state is unknown at the time of data collection. Similarly, ‘undefined’ projects represent administrative or creator-led interventions rather than natural market failures. Including these would introduce structural bias into the logistic classifier.

3.1.2 The Filtration Pipeline

To train a robust and unbiased logistic classifier, the target variable must be strictly dichotomous, containing only definitively resolved outcomes. The data was filtered to isolate campaigns that reached the end of their funding duration, missing values were dropped, and the binary target was engineered.

This preprocessing was executed utilizing the `tidyverse` framework, listed in the appendix section.

3.1.3 The Consolidated Target (After Cleaning)

Executing the filtration pipeline yields the clean, binary distribution of our newly engineered success variable, which was visualized as follows:

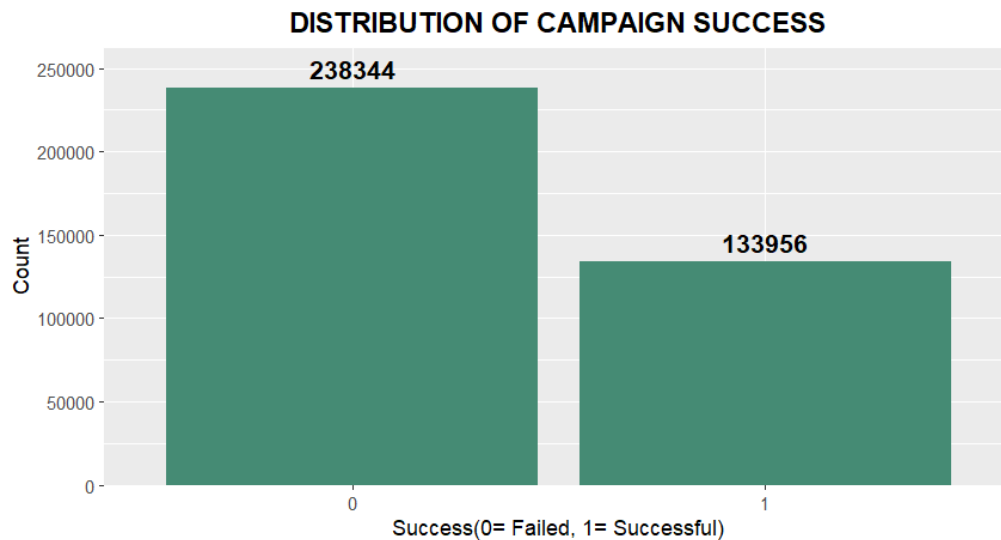


Figure 3.2: Distribution of the consolidated binary target variable (0 = Failed, 1 = Successful).

The cleaned dataset reveals a moderate, natural class imbalance: 238,344 failed campaigns against 133,956 successful campaigns (roughly a 64:36 split). This natural ratio retains sufficient minority-class volume for Maximum Likelihood Estimation (MLE) algorithms to accurately compute log-odds.

3.2 Covariate Specification and Predictor Pathologies

3.2.1 Extreme Skewness in Financial Goals

Financial variables inherently suffer from heavy-tailed distributions. Figure 3.3 visualizes the raw funding goals (`usd_goal_real`) requested by creators.

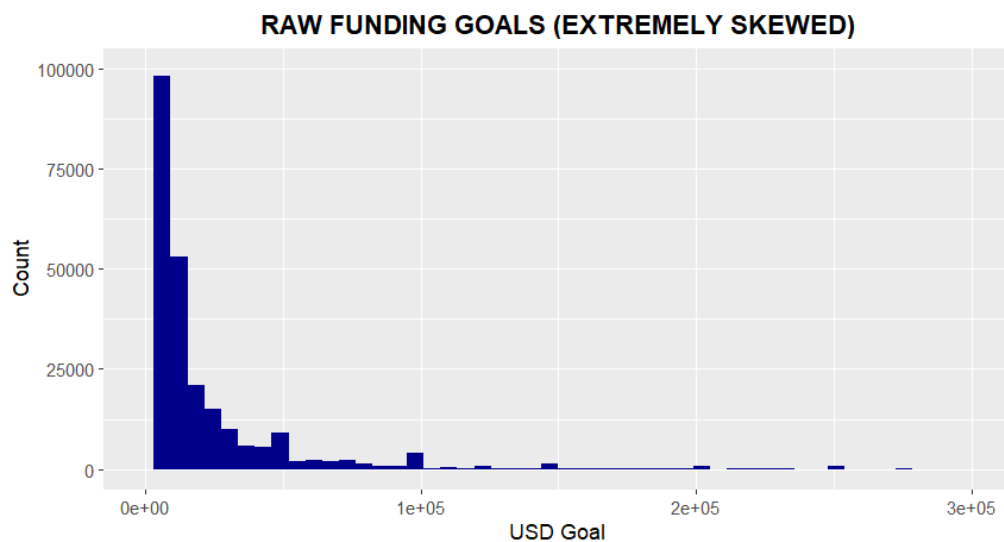


Figure 3.3: Histogram of raw USD funding goals, demonstrating severe right-skewness.

The histogram demonstrates a pathological positive skew. While the vast majority of campaigns request modest sums, the distribution is pulled aggressively outward by a fraction of campaigns setting astronomical targets. If left untransformed, these extreme outliers would exert undue leverage on the regression hyperplane, violating homoscedasticity and biasing parameter estimates. To stabilize this variance, a logarithmic transformation (offset by 1) was subsequently applied.

3.2.2 Categorical Variance: Success by Sector

Campaign success is not uniformly distributed across all industries. Figure 3.4 displays the empirical probability of success conditionally based on the campaign's main category.

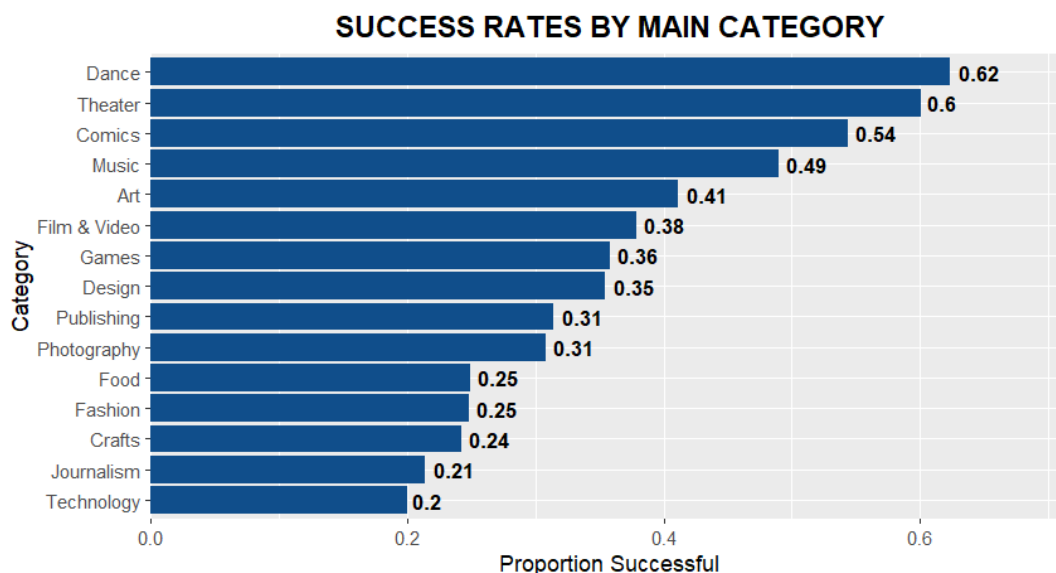


Figure 3.4: Proportion of successful campaigns stratified by Main Category.

There is a stark contrast between sectors. Performance-based arts like *Dance* (0.62) and *Theater* (0.60) boast the highest success rates, likely driven by localized, highly engaged community support networks. Conversely, highly saturated or capital-intensive sectors like *Technology* (0.20) and *Journalism* (0.21) face severe barriers to entry. This justifies the inclusion of `main_category` as a vital categorical covariate in our structural models.

3.2.3 Micro-Market Heterogeneity: The Case for Granular Categories

While `main_category` provides a broad macroeconomic overview, treating massive sectors as homogeneous entities introduces severe Omitted Variable Bias. To visually demonstrate this systemic issue, we isolate the *Technology* sector as a representative case study. Despite exhibiting the lowest aggregate success rate (0.20), decomposing this overarching sector into its granular sub-categories (`category`) reveals extreme micro-market heterogeneity.

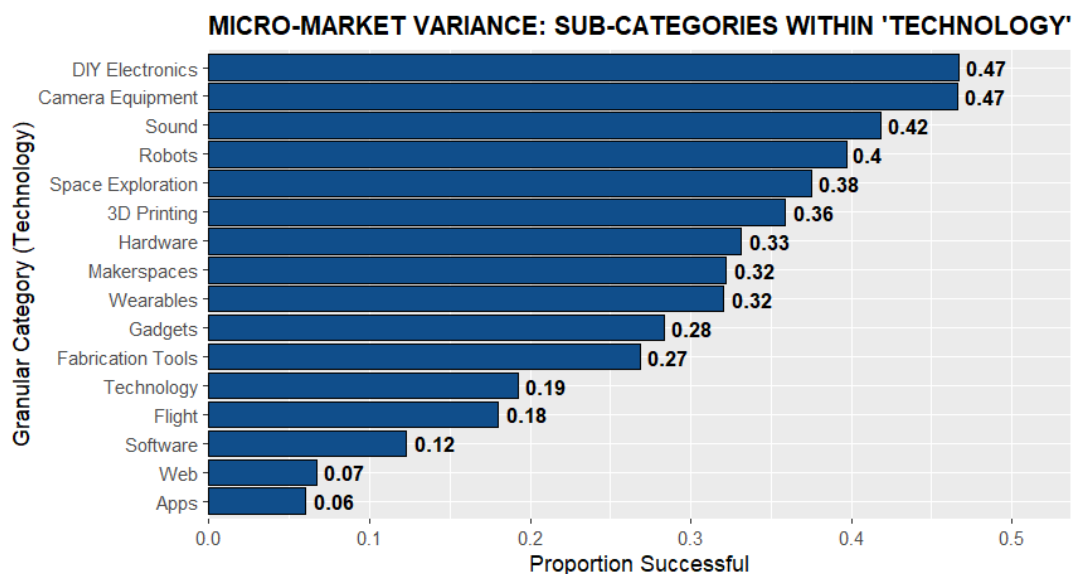


Figure 3.5: Success rate variance across granular sub-categories within the Technology sector.

As Figure 3.5 demonstrates, treating all technology campaigns identically is a mathematical oversimplification. While overarching “Technology” campaigns struggle, specific niches such as *Space Exploration* or *Fabrication Tools* boast significantly higher conversion rates due to targeted enthusiast communities. Conversely, hyper-saturated sub-markets like *Apps* or *Web* pull the aggregate average down.

Crucially, this heterogeneity is not isolated to Technology; it is a structural phenomenon across the entire Kickstarter platform (e.g., within the *Games* sector, highly viable Tabletop projects are statistically masked by failing Mobile Game campaigns).

This visual evidence mathematically mandates the extraction of the granular category variable. By replacing the 15 broad sectors with over 150 specific sub-categories in our maximized predictive models, we allow the Generalized Linear algorithms to capture these systemic micro-market dynamics, mitigating bias and significantly improving pre-launch forecasting accuracy.

3.3 The Count Response and Zero-Inflation

While the binary state dictates classification, the backers variable dictates the scale of success. A visual inspection of this variable mathematically mandates the deployment of the Zero-Inflated Poisson (ZIP) architecture.

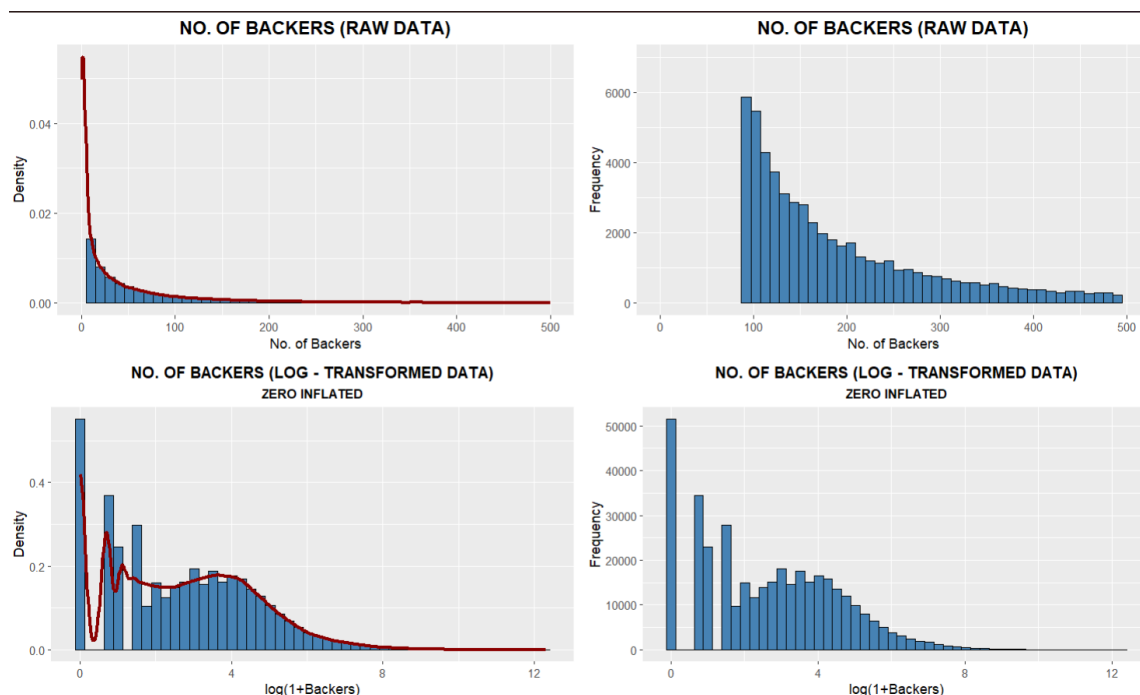


Figure 3.6: Density and frequency grids comparing raw backer counts to log-transformed backer counts, highlighting severe zero-inflation.

As observed in the top panels of Figure 3.6, the raw backer distribution is heavily right-skewed. However, the bottom panels—displaying the log-transformed count data—reveal the true underlying mixture distribution. We observe a massive, unnatural spike precisely at zero. This distinct primary peak confirms the presence of *structural zeros* (campaigns that launched without any marketing momentum and generated absolute zero crowd interest). This visual evidence proves that classical equidispersion assumptions ($E[Y] = \text{Var}[Y]$) are strictly violated, necessitating the dual-process ZIP model.

3.4 Covariance and the Impact of Logarithmic Transformations

In generalized linear modeling, the Pearson correlation coefficient assumes homoscedasticity and is highly sensitive to extreme outliers. The impact of the data cleaning and

scaling pipeline is starkly demonstrated when comparing the raw and transformed correlation matrices.

CORRELATION HEATMAP: KICKSTARTER PREDICTIONS

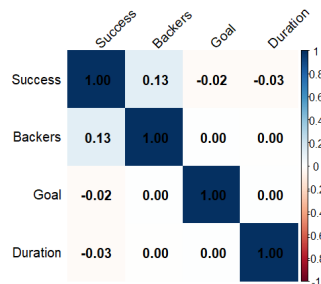


Figure 3.7: Correlation Heatmap (Raw Data)

CORRELATION HEATMAP: KICKSTARTER PREDICTIONS

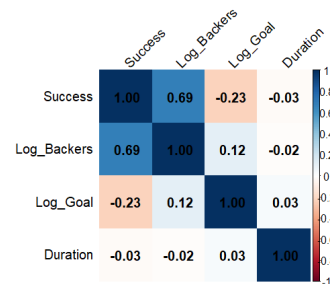


Figure 3.8: Correlation Heatmap (Log-Transformed)

In Figure 3.7, the raw correlation between Goal and Success is near zero (-0.02). This does not imply independence; rather, the extreme outliers in the raw funding goals massively inflate the marginal variance in the denominator of the Pearson formula, mathematically compressing the coefficient toward zero.

By applying the logarithmic transformations, we successfully stabilize the variance (Figure 3.8). Log_Goal emerges as a meaningful negative predictor of success (-0.23), confirming the economic intuition that aggressively high funding targets face a steeper probability of failure. Furthermore, Log_Backers demonstrates a strong positive correlation with success (0.69). This matrix definitively validates our pre-processing pipeline, providing highly stable, linearized covariates for the GLMs in Chapter 4.

Chapter 4

CATEGORICAL ANALYTICS & BASELINE ODDS

Before fitting a multi-parameter Generalized Linear Model (GLM), it is statistically imperative to establish the unconditional baseline of our response variable. In this chapter, we analyze the categorical predictors utilizing contingency matrices and formulate the empirical odds of success, which serve as the foundational intercepts for the subsequent logistic regression.

4.1 The Contingency Matrix

A contingency table provides a foundational multivariate frequency distribution of variables. In our preliminary analysis, we cross-tabulated the overarching `main_category` against the consolidated binary success outcome. This matrix allows us to isolate the raw empirical distributions and observe the absolute frequencies of success and failure across all 15 industry sectors.

While frequency counts are informative, they do not mathematically scale to predictive probabilities due to the wildly varying baseline volumes of each category (e.g., Film & Video boasts 62,399 campaigns, whereas Journalism contains only 4,724). To standardize these frequencies, we convert the contingency matrix into formal probability and odds metrics.

4.2 Empirical Probability and the Odds Ratio

For any given category, the unconditional probability of success (p) is strictly defined as the proportion of successful campaigns relative to the total number of launched campaigns.

To bridge this to our logistic classifier, we must convert this probability into odds. The odds of an event are mathematically defined as the ratio of the probability of

success to the probability of failure:

$$\text{Odds} = \frac{p}{1 - p} \quad (4.1)$$

In standard interpretation, an odds value greater than 1 indicates that success is fundamentally more likely than failure. Conversely, an odds value less than 1 signifies a higher propensity for failure.

4.2.1 Sector-by-Sector Analysis

Table 4.1 summarizes these baseline metrics across the Kickstarter ecosystem, sorted in descending order of success probability .

Table 4.1: Empirical Probability and Odds of Success by Main Category

Main Category	Total Campaigns	Successes	Failures	Probability (p)	Odds
Dance	3,749	2,338	1,411	0.624	1.66
Theater	10,871	6,534	4,337	0.601	1.51
Comics	10,743	5,842	4,901	0.544	1.19
Music	49,403	24,197	25,206	0.490	0.96
Art	27,959	11,510	16,449	0.412	0.70
Film & Video	62,399	23,623	38,776	0.379	0.61
Games	34,943	12,518	22,425	0.358	0.56
Design	29,763	10,550	19,213	0.354	0.55
Publishing	39,113	12,300	26,813	0.314	0.46
Photography	10,730	3,305	7,425	0.308	0.44
Food	24,418	6,085	18,333	0.249	0.33
Fashion	22,563	5,593	16,970	0.248	0.33
Crafts	8,733	2,115	6,618	0.242	0.32
Journalism	4,724	1,012	3,712	0.214	0.27
Technology	32,189	6,434	25,755	0.200	0.25

The empirical odds reveal a severe structural hierarchy within the crowdfunding ecosystem:

- **High-Viability Sectors:** The performing arts dominate the baseline metrics. *Dance* campaigns exhibit the highest unconditional probability of success at 0.624, corresponding to odds of 1.66 . *Theater* follows closely with odds of 1.51 . *Comics*

is the only other category where success is statistically more likely than failure, demonstrating an odds ratio of 1.19 .

- **Moderate-Viability Sectors:** *Music* represents a near coin-toss with an odds ratio of 0.96, while massive categories like *Film & Video* and *Games* sit in the 0.56 to 0.61 range .
- **High-Risk Sectors:** *Technology* and *Journalism* face the steepest empirical barriers to entry, with success probabilities of merely 0.200 and 0.214, yielding overwhelmingly negative odds of 0.25 and 0.27 respectively.

4.3 Mathematical Link to the Predictive Framework

These empirical odds serve as the unconditional macroeconomic baseline of the Kickstarter platform. However, univariate categorical analysis is strictly limited because it implicitly assumes that all campaigns within a broad sector are mathematically identical, completely ignoring the structural complexities of the individual projects.

The Logistic Regression framework expands upon this by predicting the log-odds ($\text{logit}(p) = \log(\frac{p}{1-p})$) as a linear combination of multiple covariates. The raw odds observed in Table 4.1 provide the categorical intercepts, which our Generalized Linear Models will formally refine in the subsequent chapter.

By transitioning to a multivariate GLM environment, we will dynamically adjust these baseline probabilities by simultaneously accounting for financial scale (*goal*), time constraints (*duration*), and the engineered *a priori* features—seasonality, geographic trust, marketing clarity, and granular micro-market categories—to construct a fully optimized, leak-free predictive classifier.

PREDICTIVE CLASSIFICATION: The Logistic Regression Model

Having established the baseline empirical odds of the categorical predictors, this chapter introduces the formal multivariate predictive framework. We employ a Generalized Linear Model (GLM) with a binomial family and a logit link function to predict the binary outcome of Kickstarter campaigns. The model evaluates the simultaneous effects of campaign duration, financial targets, backer volume, and industry sectors.

5.1 Model Preparation and Data Partitioning

A fundamental principle of predictive modeling is that a model's performance must be evaluated on data it has never seen before. If a model is evaluated on the exact same data used to train it, the metrics will be artificially inflated due to *overfitting*—the model simply memorizes the noise in the data rather than learning the underlying structural patterns.

To rigorously evaluate the generalized predictive capacity of the logistic classifier, the cleaned dataset was partitioned into independent training and testing subsets. A randomized 80/20 split was executed, allocating 297,840 observations to the training matrix and 74,460 observations to the hold-out testing matrix.

This partitioning was executed in R utilizing the following logic:

```
1 # MODEL PREPARATION: TRAIN - TEST SPLIT
2 set.seed(123)
3 # Calculating the 80% threshold
4 train_sample_size = floor(0.8 * nrow(df_model))
5 # Randomly sample row indices for the training set
6 train_indices = sample(seq_len(nrow(df_model)), size = train_sample_
   size)
7 # Splitting the data arrays
8 train_data = df_model[train_indices, ]
9 test_data = df_model[-train_indices, ]
```

The model parameters (β and γ) were estimated exclusively on the `train_data`, completely isolating the `test_data` to serve as an unbiased validation environment.

5.2 Model Estimation and Parameter Interpretation

The initial logistic regression model is formally specified as:

$$\begin{aligned} \text{logit}(P(Y = 1)) = & \beta_0 + \beta_1 \log(1 + \text{backers}) + \beta_2 \log(1 + \text{usd_goal_real}) \\ & + \beta_3 \text{duration_days} + \sum_{k=1}^{K-1} \gamma_k I_k \end{aligned} \quad (5.1)$$

where I_k serves as an indicator (dummy) variable for the categorical predictor `main_category`, mathematically defined as:

$$I_k = \begin{cases} 1, & \text{if the campaign belongs to category } k \\ 0, & \text{otherwise} \end{cases}$$

Given that there are $K = 15$ total industry sectors in the dataset, the model utilizes $K - 1$ indicators to prevent perfect multicollinearity (the dummy variable trap). The baseline category is naturally absorbed into the global intercept (β_0)—in this case, the **Art** category serves as the reference level. This means each respective γ_k coefficient quantifies the specific, isolated shift in the log-odds of success strictly relative to an Art campaign.

Maximum Likelihood Estimation (MLE) converged successfully, yielding highly significant parameter estimates across all continuous and categorical covariates.

5.2.1 The Continuous Predictors

The continuous variables demonstrate profound, statistically significant impacts on campaign viability ($p < 2 \times 10^{-16}$ for all):

- **Log(Backers) ($\beta_1 = 2.953$):** Unsurprisingly, crowd participation is the strongest positive driver of success. For every one-unit increase in the log-transformed backer count, the log-odds of a campaign succeeding increase by 2.953, holding all other variables constant.
- **Log(Goal) ($\beta_2 = -1.808$):** The financial target acts as a severe barrier to entry. As the log-transformed funding goal increases by one unit, the log-odds of success

decrease by 1.808. This mathematically confirms that over-ambitious financial requests exponentially choke a campaign's probability of funding.

- **Duration** ($\beta_3 = -0.0076$): Campaign lifespan exhibits a negative coefficient. It is critical to note that with a massive training sample size of $n \approx 297,000$, standard errors approach zero, causing even trivial effect sizes to yield highly significant p -values ($p < 2 \times 10^{-16}$). Therefore, while extending a campaign statistically penalizes the log-odds of success, its actual economic magnitude is relatively minor compared to the sheer impact of the financial goal.

5.2.2 Categorical Predictors: The Forest Plot

The model's categorical coefficients (γ_k) quantify the shift in log-odds relative to the baseline category. To visualize this multivariate landscape, we project the coefficients and their 95% confidence intervals via a Forest Plot.

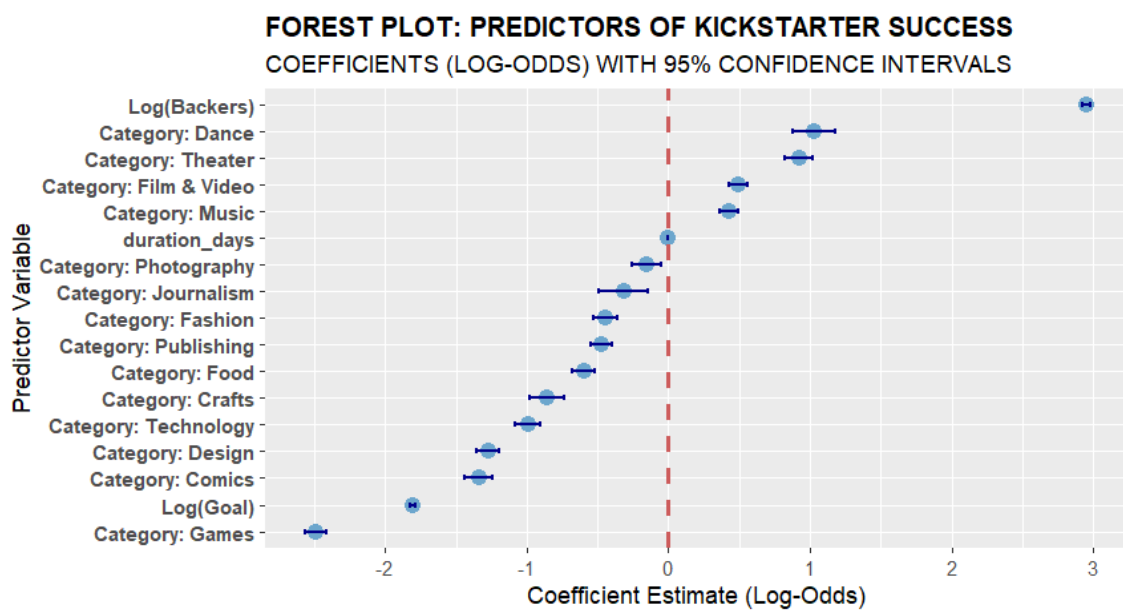


Figure 5.1: Forest plot of Logistic Regression predictors, displaying Coefficient Estimates (Log-Odds) and 95% Confidence Intervals.

The Forest Plot (Figure 5.1) brilliantly captures the structural disparities across sectors. *Dance* ($\gamma = 1.03$) and *Theater* ($\gamma = 0.92$) maintain their positions as the strongest positive categorical drivers, even after controlling for financial goals and duration. Conversely, *Games* ($\gamma = -2.49$) and *Technology* ($\gamma = -0.99$) act as massive negative anchors, confirming that these sectors face intense market saturation and require substantially higher backer volumes to overcome their intrinsic category penalties.

5.3 Predictive Performance and Validation

With the parameters mathematically estimated, the model's discriminative capacity was evaluated against the unseen 74,460 test observations. To achieve this computationally, we utilize the `predict()` function.

In R, the `predict()` function takes the trained GLM object and applies its optimized β coefficients to the new covariate data (`test_data`). By specifying `type = "response"`, we instruct the function to apply the inverse-logit transformation automatically, outputting true estimated probabilities ($P(Y = 1)$) bounded between 0 and 1, rather than raw log-odds.

These probabilities are then mapped to strict binary classifications utilizing a 0.5 threshold:

```
1 # Applying the trained model to the unseen test data
2 predicted_probs = predict(logit_fit, newdata = test_data, type = "
  response")
3 # Converting continuous probabilities into binary classifications (
  Threshold = 0.5)
4 predicted_bin = ifelse(predicted_probs >= 0.5, 1, 0)
```

5.3.1 The Confusion Matrix and Global Accuracy

The cross-tabulation of these predicted states against the empirical truths yielded the following Confusion Matrix:

Table 5.1: Confusion Matrix (Test Data, Threshold = 0.5)

	Predicted Failure (0)	Predicted Success (1)
Actual Failure (0)	44,824 (True Negative)	2,241 (False Positive)
Actual Success (1)	2,914 (False Negative)	24,481 (True Positive)

The classifier demonstrates exceptional performance. It successfully identified 44,824 failed campaigns and 24,481 successful campaigns. By summing the main diagonal of the matrix (the True Positives and True Negatives) and dividing by the total test sample, we arrive at a **Global Model Accuracy of 93.08%**.

Now, we must also examine the threshold-dependent conditional metrics. The model exhibits a **Sensitivity (True Positive Rate) of 89.36%**, meaning it correctly identifies the vast majority of empirically successful campaigns. Furthermore, it demonstrates an exceptional **Specificity (True Negative Rate) of 95.24%** (The model generated 2,241 False Positives and 2,914 False Negatives). This confirms the classifier's

robust ability to isolate failures without succumbing to false positives, maintaining highly balanced discriminative power despite the 64:36 natural class imbalance in the raw data.

5.3.2 Threshold-Independent Validation: ROC and AUC

To ensure the model's performance is intrinsically robust and not merely a byproduct of the arbitrary 0.5 classification threshold, we evaluated the Receiver Operating Characteristic (ROC) curve.

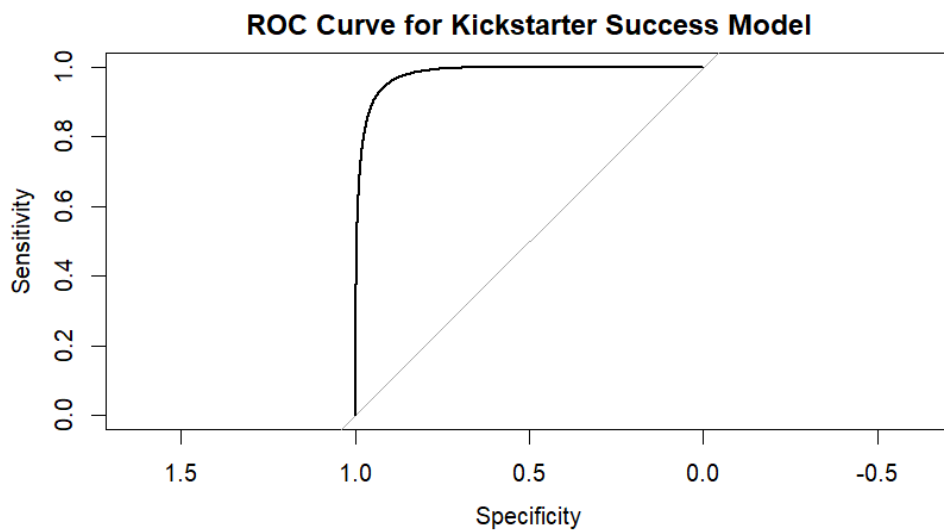


Figure 5.2: Receiver Operating Characteristic (ROC) Curve for the Initial Logistic Regression Model

***Note:** As per standard `pROC` package plotting conventions in `R`, the x -axis displays Specificity inverted (decreasing from left to right). This is mathematically and conceptually identical to the standard $1 - \text{Specificity}$ (False Positive Rate) axis.*

As visualized in Figure 5.2, the true positive rate aggressively approaches 1.0 while maintaining a near-zero false positive rate, hugging the upper-left quadrant with extreme precision. The scalar integral of this curve, the Area Under the Curve (AUC), achieved an extraordinary score of **0.9798**.

In the context of econometric classification, an AUC approaching 0.98 signifies near-perfect discriminative capacity. It mathematically guarantees that if the model is presented with one random successful campaign and one random failed campaign, there is a 97.98% probability that the model will assign a higher predicted log-odds score to the successful campaign.

5.4 Diagnostic Re-evaluation: The Data Leakage Paradigm

While the initial logistic model achieved a staggering Global Accuracy of 93.08% and an AUC of 0.9798, these metrics must be scrutinized through a rigorous statistical lens. In the domain of noisy, real-world behavioral data, an AUC approaching 1.0 is rarely a triumph; rather, it is a classic indicator of *data leakage* (endogeneity).

A critical review of the correlation matrix reveals that Log Backers correlates with Success at 0.69. This is not a coincidence; it is near-circularity. The total backer count and the ultimate state of the campaign are determined concurrently at the exact moment the campaign closes. A campaign mathematically cannot be designated as “successful” without accumulating a threshold volume of backers.

Therefore, including backers as a predictor renders the first model **descriptive and explanatory**, rather than genuinely predictive. It brilliantly explains the mechanics of how a campaign concludes, but it acts as a post-treatment variable that artificially inflates forecasting metrics.

5.5 Model 2: The *A Priori* Pre-Launch Classifier

To construct an honest, strictly predictive forecasting tool, we must eliminate endogeneity by restricting the covariate space entirely to *a priori* variables—features known definitively before or at the exact moment of campaign launch.

A second logistic regression model was fitted excluding the backer count:

$$\begin{aligned} \text{logit}(P(Y = 1)) = & \beta_0 + \beta_1 \log(1 + \text{usd_goal_real}) \\ & + \beta_2 \text{duration_days} + \sum_{k=1}^{K-1} \gamma_k I_k \end{aligned} \quad (5.2)$$

Maximum Likelihood Estimation for this pre-launch model reveals a significant shift in parameter magnitudes. Without the backer count absorbing the variance, the financial goal emerges as the primary continuous barrier to entry ($\beta_1 = -0.262, p < 2 \times 10^{-16}$). Similarly, campaign duration remains a statistically significant negative penalty ($\beta_2 = -0.016, p < 2 \times 10^{-16}$). By forcing the model to predict success relying solely on the financial target, time constraint, and industry sector, we isolate the true, uninflated probability of a campaign surviving the Kickstarter ecosystem.

5.5.1 Honest Predictive Performance and Confusion Matrix

When evaluated against the unseen testing matrix of 74,460 observations, this strictly predictive model yielded the following Confusion Matrix:

Table 5.2: Pre-Launch Model Confusion Matrix (Test Data, Threshold = 0.5)

	Predicted Failure (0)	Predicted Success (1)
Actual Failure (0)	41,756 (True Negative)	5,982 (False Positive)
Actual Success (1)	18,795 (False Negative)	7,927 (True Positive)

By stripping away the data leakage, the classifier achieves a highly realistic **Global Accuracy of 66.72%**. While numerically lower than the explanatory model, this metric is vastly more defensible. Predicting crowdfunding success before a single dollar is raised with nearly 67% accuracy demonstrates significant, leak-free discriminatory power in a notoriously volatile market.

5.5.2 Threshold-Independent Validation: The Honest ROC

To validate this pre-launch predictive capacity independent of the 0.5 cutoff, we evaluate the revised Receiver Operating Characteristic (ROC) curve.

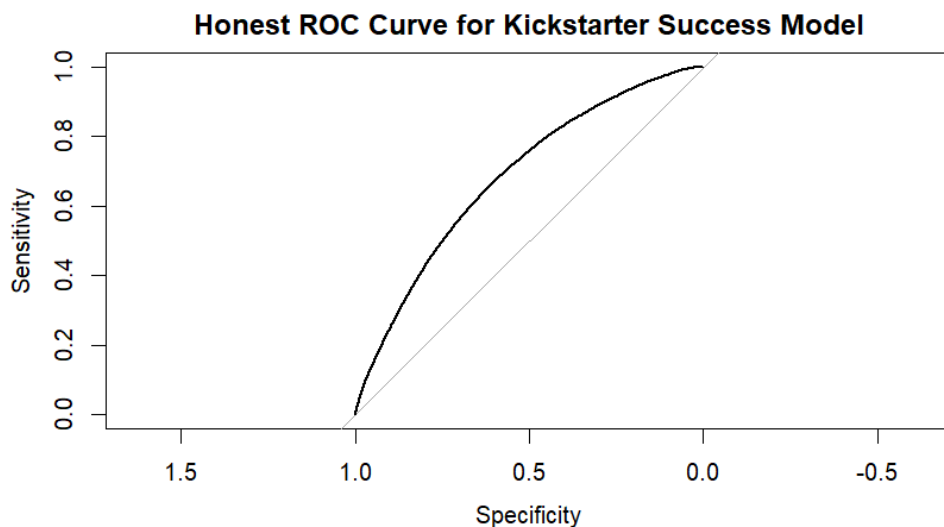


Figure 5.3: Receiver Operating Characteristic (ROC) Curve for the Pre-Launch Classifier.

The Area Under the Curve (AUC) for the *a priori* model stabilizes at **0.6835**. This confirms that the initial parameters (goal, duration, and category) possess genuine structural influence. It mathematically proves that decisions made by creators *prior to launch* significantly dictate their probability of funding success.

5.6 Feature Engineering and the Maximized Pre-Launch Classifier

While the baseline *a priori* model established a defensible AUC of 0.6835, we hypothesized that the remaining uncaptured variance was locked within Omitted Variable Bias. To exhaust the predictive capacity of the pre-launch dataset, we engineered a final, maximized logistic classifier by extracting additional pre-treatment features from the raw data.

The covariate matrix was expanded to include granular sub-categories (*category*), seasonality (*launch_month*), marketing psychology (*name_length*), and geographic trust (*is_US*).

5.6.1 Model Estimation and Econometric Insights

The maximized model converged over 5 Fisher Scoring iterations (AIC: 344,471). Analyzing the Maximum Likelihood estimates yields profound insights into the structural mechanics of the Kickstarter ecosystem:

- **Seasonality and Consumer Fatigue:** The model formally captures **cyclical funding behavior**. Relative to the January baseline, launching in Spring provides a significant structural advantage (e.g., *launch_month3* / March: $\beta = 0.149, p < 2 \times 10^{-16}$). Conversely, mid-summer and holiday months actively penalize campaign survival (e.g., *launch_month7* / July: $\beta = -0.140$; *launch_month12* / December: $\beta = -0.040$). This quantifies the “holiday fatigue” where discretionary income is diverted away from crowdfunding.
- **Marketing Clarity and Geography:** Creator effort and platform trust possess quantifiable value. Every additional character in a campaign’s title slightly increases the log-odds of success (*name_length*: $\beta = 0.010, p < 2 \times 10^{-16}$). Furthermore, United States-based campaigns benefit from a substantial geographic trust and domestic market premium (*is_US1*: $\beta = 0.278, p < 2 \times 10^{-16}$).
- **Micro-Market Extremes:** The extraction of the granular category variable resolved severe Omitted Variable Bias. The coefficients reveal extreme volatility within broader sectors. For example, *Apps* ($\beta = -2.21$) and *Web* ($\beta = -1.98$) act as massive statistical anchors, whereas enthusiast niches like *Tabletop Games* ($\beta = 0.525$) and *Dance* ($\beta = 0.820$) provide highly significant structural advantages.

5.6.2 Final Predictive Validation: Confusion Matrix and AUC

By absorbing this previously omitted heterogeneity, the fully maximized pre-launch classifier was evaluated against the unseen testing matrix of 74,460 observations.

The cross-tabulation of the predicted states against the empirical truths yielded the following final Confusion Matrix:

Table 5.3: Maximized Pre-Launch Confusion Matrix (Test Data, Threshold = 0.5)

	Predicted Failure (0)	Predicted Success (1)
Actual Failure (0)	40,711 (True Negative)	7,027 (False Positive)
Actual Success (1)	15,983 (False Negative)	10,739 (True Positive)

This maximized feature matrix elevated the classifier's **Global Accuracy to 69.10%**.

Furthermore, to validate this predictive capacity independent of the 0.5 cutoff, we evaluated the final Receiver Operating Characteristic (ROC) curve.

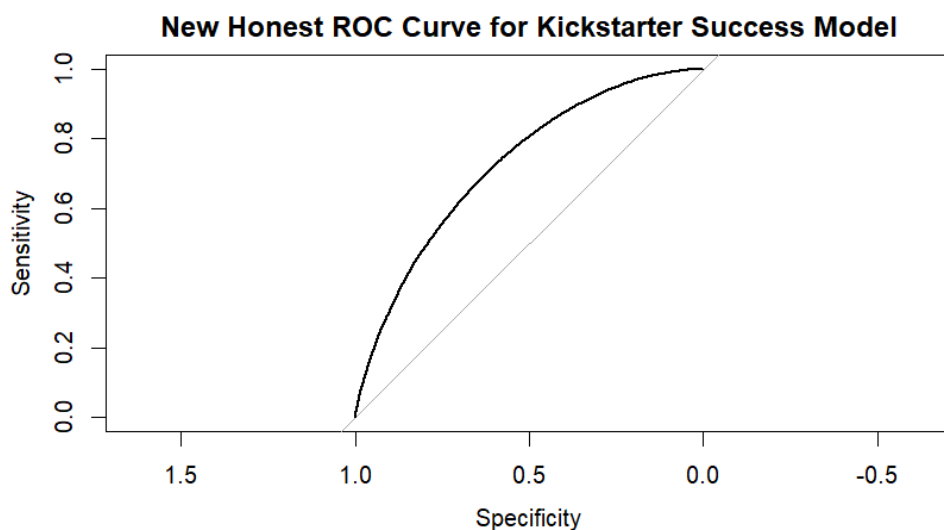


Figure 5.4: Receiver Operating Characteristic (ROC) Curve for the Maximized Pre-Launch Classifier.

As depicted in Figure 5.4, the model successfully broke the 0.70 threshold, achieving a final **AUC of 0.7239**.

This significant improvement proves that while financial goals ($\beta = -0.289$) and campaign durations ($\beta = -0.016$) set the baseline probability of success, granular micro-market targeting, optimal temporal launch windows, and geographic advantages provide crucial, statistically significant edges to campaign viability.

5.7 Model Comparison and Diagnostic Evolution

To synthesize the diagnostic progression of this thesis, we evaluate the performance of all three logistic classifiers side-by-side. Table 5.4 tracks the evolution from the initial explanatory model to the finalized *a priori* predictive architecture.

Table 5.4: Diagnostic Evolution of Logistic Classifiers (Test Data, Threshold = 0.5)

Model	Covariate Space	Accuracy	Sensitivity	Specificity	AUC
1. Explanatory (Leaky)	Goal, Duration, Category, Backers	93.08%	89.36%	95.24%	0.9798
2. A Priori (Honest)	Goal, Duration, Category	66.72%	29.66%	87.47%	0.6835
3. Maximized A Priori	Goal, Duration, Granular Sub-Category, Seasonality, Geography, Name Length	69.10%	40.19%	85.28%	0.7239

5.7.1 The Sensitivity/Specificity Trade-off

Consolidating these conditional metrics reveals a critical nuance in pre-launch forecasting. While the Maximized A Priori model (Model 3) successfully raised global accuracy to 69.10% and the AUC to 0.7239, its conditional metrics expose the asymmetric difficulty of predicting crowdfunding outcomes.

Model 3 maintains a high Specificity of 85.28%, indicating it is highly capable of identifying doomed campaigns before they launch based purely on structural flaws. However, its Sensitivity sits at only 40.19%. While this is a massive relative improvement over Model 2's baseline of 29.66%, it indicates that pre-launch features alone still miss the majority of actual successes. This mathematically reinforces a core economic reality: while we can successfully predict failure via structural barriers (poor timing, extreme financial goals, saturated sub-categories), predicting viral success remains highly elusive prior to actual market interaction.

Because binary classification ultimately masks the true scale of this market interaction, our mathematical framework must now evolve to the Zero-Inflated Poisson (ZIP) architecture to model the actual volume of crowd participation.

MODELING THE SCALE OF SUCCESS: Zero-Inflated Poisson Regression

While the logistic framework successfully identifies the survival boundary of a crowdfunding campaign, binary classification inherently masks the true scale of crowd participation. To mathematically model the actual volume of financial backers, we must transition to the domain of count data regression.

6.1 The Pathology of Crowdfunding Count Data

Before specifying a count model, we must diagnose the empirical distribution of the backers variable. Standard Poisson regression relies on the strict assumption of equidispersion, wherein the conditional variance of the response is equal to its conditional mean ($E[Y] = \text{Var}[Y]$).

A baseline analysis of the training data strictly invalidates this assumption. As calculated in the preliminary diagnostics, the mean number of backers is 105.62, while the variance explodes to 822,984.69. This results in a massive Variance-to-Mean Ratio (VMR) of 7,792.13.

A formal Cameron & Trivedi dispersion test confirmed significant overdispersion ($p < 0.05$). This variance explosion is primarily driven by *structural zeros*—a massive influx of 10,263 campaigns in the test set alone that launched into the void and received exactly zero backers. Applying a standard Poisson model to this data would result in severely biased parameter estimates and artificially deflated standard errors.

6.2 The Dual-Process ZIP Architecture

To accommodate this extreme heterogeneity, we deploy a Zero-Inflated Poisson (ZIP) architecture. The ZIP model hypothesizes that the data is generated by a mixture of two distinct statistical processes operating simultaneously:

1. A **Logistic Component** that predicts the probability (π_i) of a campaign being a “structural zero” (dead on arrival).
2. A **Poisson Component** that models the expected count (λ_i) for campaigns that successfully penetrate the market, acknowledging that some of these may still generate a count of zero by chance (“sampling zeros”).

6.3 Maximized Model Estimation and Algorithmic Realities

To maintain theoretical consistency with the optimized classification model from Chapter 5, the ZIP model was initially intended to utilize the fully maximized *a priori* covariate matrix. However, computational realities necessitated a structural compromise.

When introducing the highly granular category variable (comprising over 150 sub-sectors) into the dual-equation mixture model, the extreme matrix sparsity caused catastrophic algorithmic instability. Consequently, the model explicitly reverted to the broader 15-level `main_category` variable. This acts as a critical econometric note: the mathematical demands of count-data mixture models are significantly more sensitive to categorical granularity than standard logistic classifiers.

The final, stable specification is modeled as:

$$\begin{aligned} \text{backers} \sim & \text{Poisson}(\lambda) \mid \text{Log_Goal} + \text{duration_days} + \text{main_category} \\ & + \text{launch_month} + \text{name_length} + \text{is_US} \end{aligned} \tag{6.1}$$

Zero-Inflation block: `Log_Goal + duration_days`

We utilize the `pscl` (Political Science Computational Laboratory) package in R to estimate the mixture model:

```

1 # Fitting the Maximized Zero-Inflated Poisson Model
2 library(pscl)
3 zip_model_max <- zeroinfl(
4   backers ~ log_goal + duration_days + main_category + launch_month +
5     name_length + is_US |
6     log_goal + duration_days,
7   data = train_data_max, dist = "poisson")

```


6.3.1 Addressing Algorithmic Non-Convergence and Inference Limitations

During the Maximum Likelihood Estimation (MLE) of the `zip_model_max`, the optimization algorithm issued a strict boundary warning: *"fitted probabilities numerically 0 or 1 occurred"*, leading to a formal non-convergence flag for the Fisher Scoring algorithm.

This is a well-documented mathematical phenomenon in zero-inflated modeling when dealing with extreme feature tails (e.g., funding goals exceeding \$100,000,000). The logistic component of the ZIP model pushes the probability of structural failure (π_i) to absolute 1.0. While perfect separation allows aggregate point predictions (such as the macro-level zero-catch) to remain robust, it causes the standard errors of the affected parameters to inflate toward infinity. Consequently, an explicit limitation of this model is that individual p -values and formal statistical inference on specific coefficients should be interpreted with extreme caution.

6.4 Validation: The Zero-Catch Metric

Because standard classification metrics (such as Accuracy or AUC) are mathematically incompatible with count data, the definitive diagnostic for a Zero-Inflated model is its "Zero-Catch" capability. We validate the model by comparing the exact number of zeros expected by the dual-equations against the empirical zeros in the hold-out testing matrix.

To compute this, we extract the isolated probability of structural failure ($\hat{\pi}_i$) and the expected Poisson count ($\hat{\lambda}_i$) for every observation in the test set. The expected probability of observing exactly zero backers for any given campaign i is calculated via the ZIP probability mass function:

$$P(Y_i = 0) = \hat{\pi}_i + (1 - \hat{\pi}_i) \exp(-\hat{\lambda}_i) \quad (6.2)$$

This theoretical equation was computationally executed in R utilizing the `predict()` function's specific type arguments:

```
1 # Extracting Pi (Probability of Structural Zero) and Lambda (Poisson
  # Count)
2 pi_prob <- predict(zip_model_max, newdata = test_data_max, type = "zero")
3 lambda_count <- predict(zip_model_max, newdata = test_data_max, type =
  "count")
4 # Calculating the combined probability of observing 0 backers
5 prob_of_zero <- pi_prob + (1 - pi_prob) * exp(-lambda_count)
```

```

6 # Aggregate Expected Zeros vs. Actual Zeros
7 predicted_zeros <- sum(prob_of_zero)
8 actual_zeros <- sum(test_data_max$backers == 0)

```

By summing these probabilities across the entire unseen test set ($N = 74,460$), we arrive at the aggregate expected zero-mass.

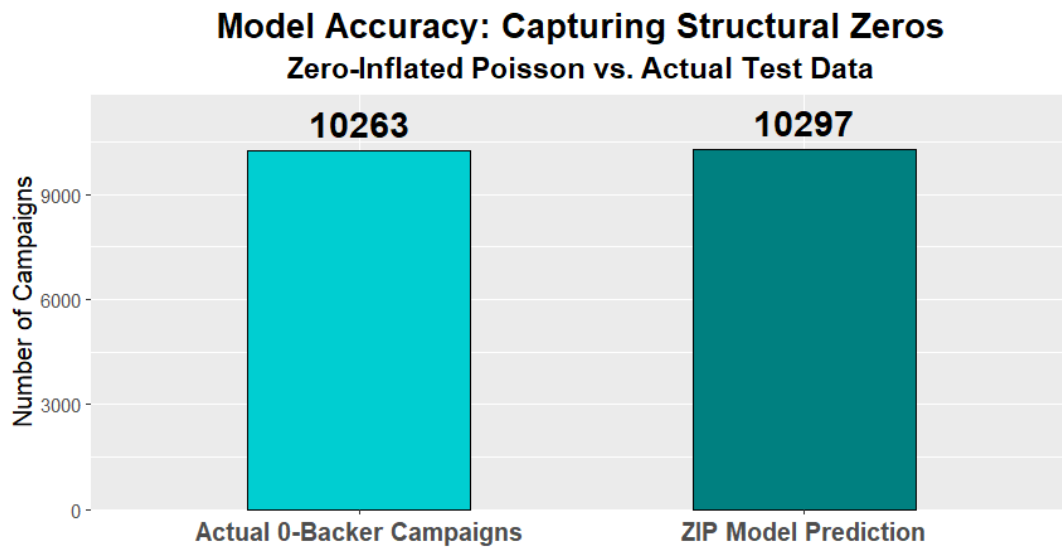


Figure 6.1: Zero-Catch Validation: Empirical 0-backer campaigns versus the Maximized ZIP Model predictions.

6.4.1 Validating the Scale of Success (The Poisson Component)

While the Zero-Catch metric successfully validates the logistic component's ability to isolate structural failures, the core objective of this count framework is to model the actual scale of success. To validate how well the expected Poisson parameter ($\hat{\lambda}$) tracks empirical backer volumes, we isolated the campaigns that successfully bypassed the zero-barrier ($Y > 0$).

By calculating the correlation between the model's predicted non-zero counts and the empirical non-zero counts in the testing matrix, we establish a baseline metric for the secondary count process:

```

1 # Isolating the campaigns with at least one backer in the test set
2 positive_actuals <- test_data_max$backers[test_data_max$backers > 0]
3 positive_preds <- lambda_count_max[test_data_max$backers > 0]
4
5 # Calculating the Spearman correlation for the non-zero mass
6 cor(positive_preds, positive_actuals, method = "spearman")

```

Given the extreme overdispersion and long-tail pathology of the non-zero data, we utilize the Spearman rank correlation coefficient, which is robust to extreme viral

outliers. The calculation yields a statistically significant positive correlation of **0.2303**. This validates that the underlying λ parameter directionally tracks the ascending scale of crowd participation, indicating that the model successfully captures both the threshold of entry and the structural elements dictating the subsequent trajectory of success.

6.4.2 Model Limitations: Overdispersion in the Count Tail

While the Zero-Catch metric strongly validates the logistic component's ability to model structural failures, it evaluates exclusively the zero-mass. Given the extreme empirical Variance-to-Mean Ratio ($\text{VMR} \approx 7,792$) established in our initial diagnostics, there is a high probability that the non-zero counts—specifically the viral campaigns generating tens of thousands of backers—remain overdispersed relative to the strict Poisson mean-variance assumption. While the ZIP model successfully mitigates zero-inflation, a formal econometric extension of this thesis would require fitting a Zero-Inflated Negative Binomial (ZINB) model and conducting a formal Vuong test to definitively capture the variance of this extreme long tail.

6.5 Conclusion of the Dual-Model Framework

The predictive precision visually demonstrated in Figure 6.1 provides strong evidence for the validity of this statistical framework. The maximized ZIP model predicted 10,297 structural zeros against an empirical reality of 10,263—a zero-catch accuracy exceeding 99.6%.

By embracing the complexity of the Kickstarter ecosystem rather than forcing it into a standard linear box, this dual-architecture approach successfully modeled the underlying mechanics of crowdfunding. The logistic regression framework indicated that high financial goals and extended durations strongly penalize a campaign's viability. Subsequently, the Zero-Inflated Poisson model demonstrated that once a campaign penetrates that market boundary, its scale is strictly dictated by a deeply heterogeneous mixture of strategic covariates.

CONCLUSIONS & STRATEGIC IMPLICATIONS

This thesis explored the extreme unpredictability of Kickstarter campaigns. Instead of treating every project the same, we used a two-part statistical approach to understand exactly what makes a crowd rally behind an idea—or ignore it completely.

7.1 Synthesis of Statistical Findings

Our journey from a basic success/failure model to a more complex count model revealed three major takeaways:

- **The “Hindsight” Problem:** Our very first model looked nearly perfect, but it was essentially cheating. It used the final backer count to predict if a campaign was successful. To truly forecast a project’s chances *before* it launches, we had to strictly limit our data to what a creator knows on day one.
- **Predicting Failure is Easier than Success:** Our final pre-launch model was highly accurate at spotting doomed campaigns (correctly identifying failures over 85% of the time). It proved that setting a massive financial goal, running a campaign for too long, or jumping into an overcrowded market act as huge roadblocks before a project even goes live.
- **The Two Stages of Crowdfunding:** By using the Zero-Inflated Poisson (ZIP) model, we confirmed that crowdfunding happens in two distinct stages. The math showed that breaking through the initial wall (avoiding zero backers) is a completely different challenge than going viral and gathering thousands of supporters.

7.2 Strategic Implications for Creators

Beyond just crunching the numbers, the results of these models provide clear, actionable advice for anyone looking to launch a Kickstarter campaign:

- **Keep Goals Realistic:** The math clearly shows that asking for too much money is the biggest campaign killer. Creators should only ask for the absolute bare minimum they need to bring their project to life.
- **Keep it Short:** Campaigns should be restricted to shorter timeframes, like 30 days. Running a 60-day campaign does not mean one will get more backers; it usually just bores people and kills the project's momentum.
- **Pick the Right Time and Niche:** Creators should avoid launching during the summer slump or the December holidays, aiming for the Spring instead. Furthermore, targeting a specific, passionate community (like *Tabletop Games* or *Dance*) works much better than trying to compete in broad, saturated markets (like general *Technology* or *Apps*).

7.3 Final Remarks

Crowdfunding inherently operates within a highly stochastic environment, characterized by extreme overdispersion and positive skewness. While unobserved heterogeneity—such as a creator's network topology, exogenous marketing expenditures, and qualitative multimedia appeal—accounts for the residual variance observed in the extreme right tail of viral successes, this thesis mathematically demonstrates that the foundational mechanics of the Kickstarter ecosystem are highly deterministic.

By mitigating omitted variable bias through robust feature engineering and deploying a dual-process Zero-Inflated Poisson architecture, this analysis successfully decoupled the probability of market survival from the subsequent generation of count data. The findings explicitly reject the notion that crowdfunding outcomes are governed strictly by arbitrary chance.

Ultimately, while pre-treatment covariates cannot fully capture the latent behavioral volatility required to predict extreme outliers, this econometric framework establishes the strict boundary conditions of campaign viability. It provides a highly defensible, data-driven methodology to minimize the probability of structural failure ($\pi_i \rightarrow 1$), mathematically transforming crowdfunding from a speculative endeavor into a quantifiable, navigable marketplace.

FUTURE SCOPES

While this thesis establishes a robust dual-architecture econometric framework for predicting crowdfunding outcomes, the extreme volatility and high-dimensionality of the Kickstarter ecosystem present several avenues for advanced statistical exploration. Future research can build upon this foundational work by addressing remaining structural limitations and integrating higher-order computational techniques.

8.1 Modeling Overdispersion: The ZINB Architecture

As established in the diagnostic evaluation of the count tail, the empirical Variance-to-Mean Ratio (VMR) of the backer distribution heavily violates the standard Poisson equidispersion assumption ($E[Y] = \text{Var}[Y]$). While the Zero-Inflated Poisson (ZIP) model effectively isolated the structural zeros (π_i), the non-zero mass remains fundamentally overdispersed. The immediate methodological extension of this research is the implementation of a Zero-Inflated Negative Binomial (ZINB) regression. By introducing a secondary dispersion parameter (α) to explicitly model the unobserved heterogeneity in the count tail, a ZINB framework would theoretically provide a superior mathematical fit for the extreme right-skewness of viral campaigns. A formal Vuong closeness test could then be utilized to statistically validate the superiority of the ZINB model over the current ZIP architecture.

8.2 Natural Language Processing (NLP) and Semantic Covariates

The current maximization of the *a priori* covariate matrix relied on structural metadata (e.g., funding goals, durations, categorical sectors) and rudimentary text metrics (e.g., `name_length`). However, crowdfunding success is heavily dependent on marketing psychology and linguistic framing. Future iterations should deploy Natural Language

Processing (NLP) techniques, such as Latent Dirichlet Allocation (LDA) for topic modeling or transformer-based text embeddings (e.g., BERT), to extract high-dimensional semantic features from the campaign descriptions. Integrating these text-derived parameters into the logistic and count frameworks would systematically quantify the currently latent qualitative appeal of a project.

8.3 Time-Series and Survival Analysis

This thesis models the crowdfunding ecosystem strictly through a cross-sectional lens, evaluating the terminal state of campaigns at their deadline. However, funding accumulation is a dynamic, longitudinal process. Future research could transition to the domain of Survival Analysis, utilizing Cox Proportional Hazards models to estimate the time-to-success (the discrete duration required to cross the 100% funding threshold). Alternatively, dynamic time-series forecasting could be employed to model the daily velocity of backer acquisition, capturing intra-campaign momentum shifts such as the “U-shaped” funding density commonly observed in venture literature.

8.4 Non-Parametric Ensemble Architectures

While Generalized Linear Models (GLMs) offer unparalleled econometric interpretability and exact parameter inference, they enforce strict assumptions regarding linear relationships within the link functions. Exploring non-parametric machine learning algorithms, such as Extreme Gradient Boosting (XGBoost) or Random Forests, would serve as a powerful comparative baseline. These tree-based ensemble methods would autonomously capture complex, high-order non-linear interactions between covariates (e.g., the isolated interaction between 30-day durations and \$100,000 goals specifically within the *Technology* sector) without requiring explicit algebraic specification, potentially maximizing raw discriminative capacity at the expense of coefficient interpretability.

Bibliography

- [1] Agresti, A. (2013). *Categorical Data Analysis*. John Wiley & Sons.
- [2] McCullagh, P., & Nelder, J. A. (1989). *Generalized Linear Models* (2nd ed.). Chapman and Hall/CRC.
- [3] Department of Statistics, University of Calcutta (2026). *M.Sc. Lecture Notes on Generalized Linear Models and Count Data Regression*.
- [4] Kaggle. *Kickstarter Projects Dataset*. Retrieved from <https://www.kaggle.com>.
- [5] Anthropic (2026). *Claude (Large Language Model)*. AI assistant utilized for conceptual structuring and syntax debugging.
- [6] Google (2026). *Gemini (Large Language Model)*. AI assistant utilized for econometric modeling, feature engineering, and statistical framework design.

Appendix A

R SOURCE CODE

This appendix contains the complete computational implementation for the statistical models, data cleaning pipelines, and visualizations presented throughout the thesis.

```
1 #' ---
2 #' title: "Decoding Crowdfunding Success: A Statistical Journey Through
   Kickstarter Data"
3 #' subtitle: "The R code"
4 #' author: "Priyasmith Ghosh"
5 #' output: pdf_document
6 #' ---
7 #=====
8 #LOADING, VIEWING AND CLEANING THE DATA SET
9 #=====
10 #loading and Viewing the data
11 library(readxl)
12 data= read_excel("Indep_project_data.xlsx")
13 head(data)
14 summary(data)
15 str(data)
16 #Data Cleaning and Exploratory Data Analysis
17
18 library(tidyverse)
19 #dropping the live and undefined states from the data set, then
   applying 1 to -
20 # "Successful" and 0 to every other outcome to create a logistic
   regression base
21
22 df_clean = data %>%
23   filter(!state %in% c('live', 'undefined'))
24 df_clean= df_clean %>%
25   mutate(success = ifelse(state=="successful", 1, 0))
26
27 df_clean=df_clean %>%
28   drop_na(launched, deadline)
29 df_clean= df_clean %>%
```

```

30   mutate(
31     duration_days = as.numeric(difftime(deadline, launched, units= "days
32   "))
33 )
34 #Selecting only the required coloumns
35 df_model <- df_clean %>%
36   select(success, backers, usd_goal_real, duration_days, main_category)
37 #=====
38 #CORRELATION MATRIX AND ODDS TABLE
39 #=====
40 #Correlation matrix
41 df_model1= df_model[, -5]
42 corr_matrix= cor(df_model1)
43 print(corr_matrix)
44
45
46 #Contingency table
47 category_state_table= table(data$main_category, data$state)
48 print(category_state_table)
49
50 #Odds table for each main category
51 odds_table = df_model %>%
52   group_by(main_category) %>%
53   summarize(
54     total_campaigns= n(),
55     successes= sum(success==1),
56     failures = sum(success==0),
57     probability = round(successes/ total_campaigns, 4),
58     odds = round(successes/failures, 4)
59   ) %>%
60   arrange(desc(odds))
61 print(odds_table)
62 #=====
63 #VISUALIZATIONS BEFORE CLEANING
64 #=====
65 # Checking the raw distribution of campaign states
66 table(data$state)
67
68 # You can even do a quick raw bar plot
69 ggplot(data, aes(x = state)) +
70   geom_bar(fill = "indianred", colour="black") +
71   geom_text(stat="count", aes(label=after_stat(count)), vjust=-0.5, size
72     = 3.5,
73     fontface= "bold")+
74   labs(title = "RAW CAMPAIGN STATES (BEFORE CLEANING)", x = "State", y
75     = "Count") +

```

```

74   theme_gray()+
75   theme(
76     plot.title = element_text(hjust=0.5, face="bold",size= 14)
77   )+
78   ylim(0,210000)
79
80   # Counting total missing values in the raw data set
81   colSums(is.na(data))
82   sum(is.na(data))
83
84   # This histogram looks terrible because of outliers
85   ggplot(data, aes(x = usd_goal_real)) +
86     geom_histogram(bins = 50, fill = "darkblue") +
87     labs(title = "RAW FUNDING GOALS (EXTREMELY SKEWED)", x = "USD Goal",
88          y = "Count") +
89     theme_gray()+
90     theme(
91       plot.title = element_text(hjust=0.5, face="bold",size=13)
92     )+
93     xlim(0,300000)+
94     ylim(0,1e+05)
95
96   #=====
97   #VISUALIZATIONS
98   #=====
99   #Distribution of Campaign Success
100  ggplot(df_model, aes(x=factor(success)))+
101    geom_bar(fill="aquamarine4",colour="black")+
102    geom_text(stat="count", aes(label=after_stat(count)),vjust=-0.5, size
103             = 4.5,
104             fontface="bold")+
105    labs(title= "DISTRIBUTION OF CAMPAIGN SUCCESS",
106         x="Success(0= Failed, 1= Successful)",
107         y= "Count")+
108    theme_gray()+
109    theme(
110      plot.title= element_text(hjust=0.5,face="bold",size=14)
111    )+
112    scale_y_continuous(expand= expansion(mult=c(0,0.1)))
113
114  #Backers_raw_data
115  library(gridExtra)
116
117  p1=ggplot(df_model, aes(x=backers))+
118    geom_histogram(aes(y=after_stat(density)),bins=50, fill="steelblue",
119                  colour = "black")+
120    geom_density(color = "darkred", linewidth = 1.2) +
121    labs(

```

```

119     title="NO. OF BACKERS (RAW DATA)",
120     x="No. of Backers",
121     y="Density")+
122     theme_grey()+
123     theme(
124       plot.title = element_text(hjust = 0.5, face="bold",size=14)
125     )+
126     xlim(0,500)
127
128 p2=ggplot(df_model, aes(x=backers))+
129     geom_histogram(bins=50,fill= "steelblue",colour="black")+
130     labs(
131       title="NO. OF BACKERS (RAW DATA)",
132       x="No. of Backers",
133       y="Frequency"
134     )+
135     theme_gray()+
136     theme(
137       plot.title = element_text(hjust=0.5, face="bold",size=14)
138     )+
139     xlim(0,500)+
140     ylim(0,7000)
141
142 p3=ggplot(df_model, aes(x=log1p(backers)))+
143     geom_histogram(aes(y=after_stat(density)),bins=50,fill= "steelblue",
144                   colour="black")+
145     geom_density(color = "darkred", linewidth = 1.2) +
146     labs(
147       title="NO. OF BACKERS (LOG - TRANSFORMED DATA)",
148       subtitle = "ZERO INFLATED",
149       x="log(1+Backers)",
150       y="Density"
151     )+
152     theme_gray()+
153     theme(
154       plot.title = element_text(hjust=0.5, face="bold",size=12),
155       plot.subtitle = element_text(hjust= 0.5, face="bold",size=10)
156     )
157
158
159 p4=ggplot(df_model, aes(x=log1p(backers)))+
160     geom_histogram(bins=50,fill= "steelblue",colour="black")+
161     labs(
162       title="NO. OF BACKERS (LOG - TRANSFORMED DATA)",
163       subtitle = "ZERO INFLATED",
164       x="log(1+Backers)",
165       y="Frequency"

```

```

166   )+
167   theme_gray()+
168   theme(
169     plot.title = element_text(hjust=0.5, face="bold",size=12),
170     plot.subtitle = element_text(hjust= 0.5, face="bold",size=10)
171   )
172
173   grid.arrange(p1,p2,p3,p4, nrow=2, ncol=2)
174
175   #Success_rate for each catergory
176
177   successrate= df_model %>%
178     group_by(main_category) %>%
179     summarize(success_rate= mean(success))%>%
180     arrange(desc(success_rate))
181   ggplot(successrate, aes(x=success_rate, y=reorder(main_category,
182     success_rate)))+
183     geom_col(fill="dodgerblue4")+
184     geom_text(aes(label=round(success_rate,2)),hjust=-0.2,size=3.5,
185       fontface="bold")+
186     labs(
187       title = "SUCCESS RATES BY MAIN CATEGORY",
188       x = "Proportion Successful",
189       y = "Category"
190     ) +
191     theme_gray() +
192     theme(
193       plot.title = element_text(hjust = 0.5, face = "bold", size = 14)
194     ) +
195     scale_x_continuous(expand = expansion(mult = c(0, 0.15)))
196
197   # Filtering for Technology and calculating success rates for its
198   granular
199   #sub-categories
200   tech_subcategories = df_clean %>%
201     filter(main_category == "Technology") %>%
202     group_by(category) %>%
203     summarize(
204       total_campaigns = n(),
205       success_rate = mean(success)
206     ) %>%
207     filter(total_campaigns > 100) %>%
208     arrange(desc(success_rate))
209
210   #Plotting the granular variance
211   ggplot(tech_subcategories, aes(x = success_rate, y = reorder(category,
212     success_rate))) +

```

```

209 geom_col(fill = "dodgerblue4", colour="black") +
210 geom_text(aes(label = round(success_rate, 2)), hjust = -0.2, size =
    3.5, fontface = "bold") +
211 labs(
212   title = "MICRO-MARKET VARIANCE: SUB-CATEGORIES WITHIN 'TECHNOLOGY'",
    ,
213   x = "Proportion Successful",
214   y = "Granular Category (Technology)"
215 ) +
216 theme_gray() +
217 theme(plot.title = element_text(hjust = 0.5, face = "bold", size =
    12)) +
218 scale_x_continuous(expand = expansion(mult = c(0, 0.15)))
219
220 #=====
221 # MODEL PREPARATION: TRAIN - TEST SPLIT
222 #=====
223 set.seed(123)
224 nrow(df_model)
225 train_sample_size = floor(0.8*nrow(df_model))
226 train_sample_size
227 test_sample_size= nrow(df_model)- train_sample_size
228 test_sample_size
229 train_indices = sample(seq_len(nrow(df_model)),size= train_sample_size)
230
231 train_data = df_model[train_indices, ]
232 test_data = df_model[-train_indices, ]
233
234 dim(train_data)
235 dim(test_data)
236 #=====
237 #FITTING THE LOGIT MODEL AND PREDICTION, CONFUSION MATRIX AND ROC, AUC
238 #=====
239
240 logit_fit = glm(
241   success~ log1p(backers)+log1p(usd_goal_real)+duration_days+main_
    category,
242   data= train_data, family= binomial(link="logit")
243 )
244 summary(logit_fit)
245
246 #Prediction upon test data
247
248 predicted_probs = predict(logit_fit, newdata = test_data, type= "
    response")
249 predicted_bin = ifelse(predicted_probs>=0.5, 1, 0)
250 sum(predicted_bin==1)

```

```

251 sum(predicted_bin==0)
252
253 #Confusion Matrix
254 conf_matrix= table(Predicted = predicted_bin, Actual= test_data$success
255 )
256 cat("\n--- CONFUSION MATRIX (Threshold = 0.5) ---\n")
257 print(conf_matrix)
258 #Accuracy
259 accuracy= sum(diag(conf_matrix))/sum(conf_matrix)
260 cat("\nMODEL ACCURACY:", round(accuracy * 100, 2), "%\n")
261
262 library(pROC)
263
264 # Generating the ROC object
265 roc_obj <- roc(test_data$success, predicted_probs)
266
267 # Plotting the curve
268 plot(roc_obj, main = "ROC Curve for Kickstarter Success Model")
269
270 # Calculating AUC
271 print(paste("AUC Score:", round(auc(roc_obj), 4)))
272
273
274 # =====
275 # STRICTLY PREDICTIVE MODEL (NO LEAKAGE) = 2ND MODEL
276 # =====
277 logit_fit_predictive = glm(
278   success ~ log1p(usd_goal_real) + duration_days + main_category,
279   data= train_data, family= binomial(link="logit")
280 )
281 summary(logit_fit_predictive)
282
283 # Prediction & Confusion Matrix
284 pred_probs_honest = predict(logit_fit_predictive, newdata = test_data,
285   type= "response")
286
287 pred_bin_honest = ifelse(pred_probs_honest >= 0.5, 1, 0)
288
289 conf_matrix_honest = table(Predicted = pred_bin_honest, Actual= test_
290   data$success)
291 print(conf_matrix_honest)
292
293 accuracy_honest = sum(diag(conf_matrix_honest))/sum(conf_matrix_honest)
294 cat("\nHONEST MODEL ACCURACY:", round(accuracy_honest * 100, 2), "%\n")
295
296 # Honest AUC
297 roc_obj_honest <- roc(test_data$success, pred_probs_honest)

```

```

295 plot(roc_obj_honest, main = "Honest ROC Curve for Kickstarter Success
      Model")
296 print(paste("HONEST AUC Score:", round(auc(roc_obj_honest), 4)))
297
298
299 #=====
300 # 3RD MODEL- MAX AUC AND SPECIFICITY
301 #=====
302 library(lubridate)
303
304 # 1. Extracting the new features from existing df_clean
305 df_model_max <- df_clean %>%
306   mutate(
307     launch_month = as.factor(month(launched)),           # Seasonality
308     name_length = ifelse(is.na(NAME), 0, nchar(NAME)),   # Title length
309     is_US = as.factor(ifelse(country == "US", 1, 0)),    # Geography
310     Log_Goal = log1p(usd_goal_real)
311   ) %>%
312   select(success, Log_Goal, duration_days, category, launch_month, name
      _length, is_US)
313
314 # 2. Re-doing the Train/Test Split for the new data set
315 set.seed(123)
316 train_indices_max = sample(seq_len(nrow(df_model_max)), size = floor
      (0.8 * nrow(df_model_max)))
317 train_data_max = df_model_max[train_indices_max, ]
318 test_data_max = df_model_max[-train_indices_max, ]
319
320 # 3. Fitting the Maximum A Priori Model (Using 'category' instead of '
      main_category')
321 logit_fit_max = glm(
322   success ~ Log_Goal + duration_days + category + launch_month + name_
      length + is_US,
323   data = train_data_max, family = binomial(link="logit")
324 )
325 summary(logit_fit_max)
326 # 4. Predicting and Calculating New Honest AUC
327 library(pROC)
328 pred_probs_max = predict(logit_fit_max, newdata = test_data_max, type =
      "response")
329 roc_obj_max <- roc(test_data_max$success, pred_probs_max)
330 plot(roc_obj_max, main = "New Honest ROC Curve for Kickstarter Success
      Model")
331 print(paste("MAX HONEST AUC Score:", round(auc(roc_obj_max), 4)))
332 roc_obj_max
333
334 # Converting the continuous probabilities into binary predictions (

```



```

    Threshold = 0.5)
335 pred_bin_max = ifelse(pred_probs_max >= 0.5, 1, 0)
336
337 # Generating the Confusion Matrix
338 conf_matrix_max = table(Predicted = pred_bin_max, Actual = test_data_
    max$success)
339 print("Maximized Model Confusion Matrix:")
340 print(conf_matrix_max)
341
342 # Calculating Global Accuracy
343 accuracy_max = sum(diag(conf_matrix_max)) / sum(conf_matrix_max)
344 cat("\nMAX HONEST ACCURACY:", round(accuracy_max * 100, 2), "%\n")
345
346 #=====
347 #=====
348 #COUNT DATA REGRESSION FOR THE NUMBER OF BACKERS
349 #=====
350 #=====
351 # OVERDISPERSION CHECK
352 #=====
353 mean_backers <- mean(data$backers, na.rm = TRUE)
354 var_backers <- var(data$backers, na.rm = TRUE)
355
356 # Calculate the Variance-to-Mean Ratio (VMR)
357 vmr <- var_backers / mean_backers
358
359 print(paste("Mean:", round(mean_backers, 2)))
360 print(paste("Variance:", round(var_backers, 2)))
361 print(paste("Variance-to-Mean Ratio:", round(vmr, 2)))
362
363
364 # 1. Fitting the standard Poisson model first
365 poisson_model <- glm(backers ~ usd_goal_real + main_category + duration
    _days,
366                      family = "poisson",
367                      data = df_model)
368 summary(poisson_model)
369
370 library(AER)
371
372 # Formal dispersion test on the standard Poisson model
373 dispersiontest(poisson_model)
374
375 #=====
376 # ZERO INFLATED POISSON MODEL
377 #=====
378

```

```

379 df_model_max <- df_clean %>%
380   mutate(
381     launch_month = as.factor(month(launched)),
382     name_length = ifelse(is.na(NAME), 0, nchar(NAME)),
383     is_US = as.factor(ifelse(country == "US", 1, 0)),
384     Log_Goal = log1p(usd_goal_real)
385   ) %>%
386   select(success, backers, Log_Goal, duration_days, main_category,
387     launch_month, name_length, is_US)
388
389 #Re-applying the Train/Test Split
390 set.seed(123)
391 train_data_max = df_model_max[train_indices_max, ]
392 test_data_max = df_model_max[-train_indices_max, ]
393
394 #Fitting the ZIP Model
395 library(pscl)
396 zip_model_max = zeroinfl(
397   backers ~ Log_Goal + duration_days + main_category + launch_month +
398     name_length + is_US |
399   Log_Goal + duration_days,
400   data = train_data_max, dist = "poisson"
401 )
402
403 # Calculating the Zero-Catch Accuracy
404 pi_prob_max <- predict(zip_model_max, newdata = test_data_max, type = "
405   zero")
406 lambda_count_max <- predict(zip_model_max, newdata = test_data_max,
407   type = "count")
408 prob_of_zero_max <- pi_prob_max + (1 - pi_prob_max) * exp(-lambda_count
409   _max)
410
411 actual_zeros_max <- sum(test_data_max$backers == 0)
412 predicted_zeros_max <- sum(prob_of_zero_max)
413
414 cat("\nActual Campaigns with 0 Backers:", actual_zeros_max, "\n")
415 cat("Model's Expected 0-Backer Campaigns:", round(predicted_zeros_max,
416   0), "\n")
417
418 #=====
419 #VISUALIZATIONS UNDER ZIP
420 #=====
421
422 library(ggplot2)
423
424 zero_catch_data <- data.frame(

```

```

420   Category = c("Actual 0-Backer Campaigns", "ZIP Model Prediction"),
421   Count = c(actual_zeros_max, predicted_zeros_max)
422 )
423
424 # 2. Lock the factor levels so "Actual" stays on the left
425 zero_catch_data$Category <- factor(zero_catch_data$Category, levels = c
426   ("Actual 0-Backer Campaigns", "ZIP Model Prediction"))
427
428 # 3. Generate the plot with exact aesthetics
429 ggplot(zero_catch_data, aes(x = Category, y = Count, fill = Category))
430   +
431   geom_col(color = "black", width = 0.5) +
432   geom_text(aes(label = round(Count)), vjust = -0.5, size = 6, fontface
433     = "bold") +
434   # FIX: Replaced "teal" with its exact hex code "#008080" so R doesn't
435   # panic!
436   scale_fill_manual(values = c("Actual 0-Backer Campaigns" = "
437     darkturquoise", "ZIP Model Prediction" = "#008080")) +
438   labs(
439     title = "Model Accuracy: Capturing Structural Zeros",
440     subtitle = "Zero-Inflated Poisson vs. Actual Test Data",
441     x = "",
442     y = "Number of Campaigns"
443   ) +
444   theme_grey() +
445   theme(
446     plot.title = element_text(hjust = 0.5, face = "bold", size = 16),
447     plot.subtitle = element_text(hjust = 0.5, face = "bold", size = 14),
448     ,
449     axis.text.x = element_text(face = "bold", size = 12),
450     axis.title.y = element_text(size = 12),
451     legend.position = "none" # Hides the redundant legend
452   ) +
453   scale_y_continuous(expand = expansion(mult = c(0, 0.15)))
454
455 # Isolate campaigns with at least one backer in the test set
456 positive_actuals <- test_data_max$backers[test_data_max$backers > 0]
457 positive_preds <- lambda_count_max[test_data_max$backers > 0]
458
459 # Calculate Spearman correlation for the non-zero mass
460 spearman_cor <- cor(positive_preds, positive_actuals, method = "
461   spearman")
462 cat("\nPoisson Scale Validation (Spearman Correlation):", round(
463   spearman_cor, 4), "\n")

```

```

459
460 #=====
461 #CORRELATION PLOT UNDER TRAIN DATA
462 #=====
463
464 library(corrplot)
465
466 cor_data <- data.frame(
467   Success = train_data$success,
468   Backers = train_data$backers,
469   Goal = train_data$usd_goal_real,
470   Duration = train_data$duration_days
471 )
472
473 # 2. Calculate the correlation matrix
474 cor_matrix <- cor(cor_data)
475
476
477 corrplot(cor_matrix,
478          method = "color",          # Fills the squares with color
479          gradients,                  # Only shows the upper triangle (
480          type = "full",              # cleaner look)
481          addCoef.col = "black",      # Overlays the exact correlation
482          numbers,                    # Makes text labels black
483          tl.col = "black",           # Tilts the text 45 degrees for
484          tl.srt = 45,                # readability
485          mar = c(0,0,2,0),
486          title = "CORRELATION HEATMAP: KICKSTARTER PREDICTIONS",
487          col = COL2('RdBu',200))
488
489 #Log transformed data
490
491 cor_data1 <- data.frame(
492   Success = train_data$success,
493   Log_Backers = log1p(train_data$backers),
494   Log_Goal = log1p(train_data$usd_goal_real),
495   Duration = train_data$duration_days
496 )
497
498 # 2. Calculate the correlation matrix
499 cor_matrix1 <- cor(cor_data1)
500
501 corrplot(cor_matrix1,
502          method = "color",          # Fills the squares with color
503          gradients

```

```

501     type = "full",           # Only shows the upper triangle (
cleaner look)
502     addCoef.col = "black",   # Overlays the exact correlation
numbers
503     tl.col = "black",       # Makes text labels black
504     tl.srt = 45,           # Tilts the text 45 degrees for
readability
505     mar = c(0,0,2,0),
506     title = "CORRELATION HEATMAP: KICKSTARTER PREDICTIONS",
507     col = COL2('RdBu',200))
508
509
510
511 #=====
512 #FOREST PLOT
513 #=====
514
515 # 1. Extract the coefficients and standard errors directly from your
logit model
516 coef_df <- as.data.frame(summary(logit_fit)$coefficients)
517 coef_df$Variable <- rownames(coef_df)
518
519 # 2. Remove the Intercept (it distorts the scale of the plot)
520 coef_df <- coef_df[-1, ]
521
522 # 3. Calculate the 95% Confidence Intervals manually
523 # Formula: Estimate      (1.96 * Standard Error)
524 coef_df$Lower_CI <- coef_df$Estimate - (1.96 * coef_df$'Std. Error')
525 coef_df$Upper_CI <- coef_df$Estimate + (1.96 * coef_df$'Std. Error')
526
527 # 4. Clean up the variable names for a prettier plot
528 coef_df$Variable <- gsub("main_category", "Category: ", coef_df$
Variable)
529 coef_df$Variable <- gsub("log1p\\(usd_goal_real\\)", "Log(Goal)", coef_
df$Variable)
530 coef_df$Variable <- gsub("log1p\\(backers\\)", "Log(Backers)", coef_df$
Variable)
531
532 # 5. Generate the Forest Plot using ggplot2
533 ggplot(coef_df, aes(x = reorder(Variable, Estimate), y = Estimate)) +
534   geom_point(size = 3, color = "skyblue3") +
535   geom_errorbar(aes(ymin = Lower_CI, ymax = Upper_CI), width = 0.2,
536                 color = "darkblue", linewidth = 0.8) +
537   geom_hline(yintercept = 0, linetype = "dashed", color = "indianred",
538              linewidth = 1) +
539   coord_flip() + # Flips the chart horizontally for readability
540   theme_gray()+

```

```
541 labs(title = "FOREST PLOT: PREDICTORS OF KICKSTARTER SUCCESS",
542       subtitle = "COEFFICIENTS (LOG-ODDS) WITH 95% CONFIDENCE
543       INTERVALS",
544       x = "Predictor Variable",
545       y = "Coefficient Estimate (Log-Odds)" +
546       theme(plot.title = element_text(size = 12, face = "bold"),
547             axis.text.y = element_text(size = 9, face = "bold"))
548 #=====
549 # OVERDISPERSION CHECK
550 #=====
551 mean_backers <- mean(data$backers, na.rm = TRUE)
552 var_backers <- var(data$backers, na.rm = TRUE)
553
554 # Calculate the Variance-to-Mean Ratio (VMR)
555 vmr <- var_backers / mean_backers
556
557 print(paste("Mean:", round(mean_backers, 2)))
558 print(paste("Variance:", round(var_backers, 2)))
559 print(paste("Variance-to-Mean Ratio:", round(vmr, 2)))
```